The Contained Book: An Absorbing Resource Measured to Be Worth Nothing

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Abstract

In the six-player partnership game Literature (Canadian Fish), a half-suit whose six cards all sit with one team is an *absorbing state*: the opponents can neither ask into it nor take it by declaring, so leaving it unclaimed is a resource rather than an oversight. The resource is concrete — a holder may deliberately ask into the set, which is a guaranteed miss, and a miss hands the turn to an opponent of the asker's choosing without moving a card. This yields a renewable, aimable, nearly information-free turn-pass that only the containing team can play and that claiming the set destroys. We prove the absorbing property from the rules of both rule sets the FishAI engine supports — correcting, along the way, the project's own earlier claim that it failed under the 48-card baseline — derive a trigger inequality that prices the move in cards against the ordinary ask it would displace, implement it as a policy option with a strict no-convention discipline, and measure it at three scales: per-decision mirrors, 1,200 duplicate-deal pairs per style, and a full 36-cell × 4,300-pair round-robin re-run. The mechanism fires, fires for exactly the reasons the derivation gives, aims where the style asks, and reuses one card so that 95–98% of its uses publish nothing new. Its measured value is zero: no style moves two standard errors from 0.5 in the paired mirror, and the full matrix re-run leaves the verdict, the ranking, and the cyclic structure unchanged, with the heaviest intended user shifting by -0.0018 ± 0.0014. The arithmetic says why: at the roster's measured hit rate a conceded turn is worth about one card, so the entire spread between the best and worst opponent to concede to is about one card. We report this as a model negative result — about the move at this level of play, not about the theorem.

1 Introduction

Negative results are cheap to assert and expensive to earn. This paper earns one. It concerns a single strategic mechanism in Literature (Canadian Fish) under the FishAI project's `us54` dialect: the *contained book*, a half-suit wholly held by one team. The mechanism has three parts, each of which survives scrutiny — and a value, which does not.

The parts. First, a theorem: containment is absorbing. No opponent can legally ask into a contained book, and an opponent who declares it hands it to the holders, so the position can only be ended by the holders themselves (Section 3). Second, a move: a holder may ask into the book anyway. The ask is legal, cannot hit, and on the miss the turn passes to whichever opponent the asker named — a renewable, aimable turn-pass that costs at most one card of hand information over its lifetime if one card is reused (Section 4). Third, a price: a derived inequality, with every term measured off public data in units of cards, that says when the pass beats the ordinary ask it would displace (Section 5).

The value. Implemented with that pricing and handed to a nine-style roster, the move is worth nothing. Across 1,200 duplicate-deal pairs per style, no style equipped with the move beats its own unequipped twin by two standard errors (Section 6.3). Across a full-precision re-run of the project's 36-cell round-robin [1] — same 4,300 seeds, same rotation, the policy change and nothing else — the verdict, the ranking, and the cyclic structure are all unchanged (Section 6.4). The measured value of the contained-book turn-pass, as a policy option under `us54` against the roster it was built for, is zero [3].

Two features make this negative result worth writing up rather than filing away. The first is that every stage of it is *measured rather than asserted*, and twice the measurement corrected the project's own prose: the handoff's claim that the move is information-free (it is not, but the qualification makes the move better — Section 4); and an earlier draft's claim that containment is not absorbing under the 48-card baseline rules (it is — Section 3.1). The second is that the mechanism demonstrably works. It fires at the derived rate, refuses in the derived cases, and never leaks — so its worthlessness is a finding about the game at this level of play, not a bug report. The arithmetic bounding the effect at about one card (Section 7) tells us where value would have to come from if it exists at all: the declare policy, which is untouched here (Section 8).

2 The rules the result rests on

Literature is played by six players in two teams of three (seats $0, 2, 4$ versus $1, 3, 5$) [8]. Under `us54` [4] the deck is 54 cards — the standard 52 plus two distinguishable jokers — partitioned into nine *books* (half-suits) of six cards: per suit, LOW = 2–7 and HIGH = 9–A, plus an EIGHTS book of the four 8s and both jokers. Nine cards are dealt to each player; the game ends the moment either team has been awarded five books. The project's shipped baseline, `pagat48` [5], is the 48-card game (8s removed, eight books, play to the end).

The theorem needs only a handful of rows from the `us54` decision table [4]; we reproduce them by number, since the proof cites them.

row	rule
5	**Ask target.** The target of an ask is always an opponent; a teammate may never be asked.
6	**Ask licence.** An asker must hold at least one card of the asked book.
7	**Own card.** You may not ask for a card you hold.
9	**Hit.** The card transfers; the asker keeps the turn and asks again.
10	**Miss.** The turn passes to the player who was asked.
12	**Declare content.** Name the book and the exact holder of all six cards; every assignment must be a seat on the declarer's own team.
14	**Any error at all.** The opposing team scores the book. Under `us54` this covers both "an opponent held one of the six" and "my team held all six but I misassigned"; the `pagat48` *void* outcome is abolished.
15	**Declare without holding.** Legal — you may declare a book you hold no card of.
17	**Information.** Card counts are public, with a full public log of asks, results, and declares.

Under `pagat48` the corresponding declare rows split the error case: its row 14 awards the book to the opposing team when an opponent of the declarer holds at least one card, regardless of other locations, and its row 15 *voids* the book when the declarer's own team held all six but a location was wrong [5]. The distinction matters in Section 3.1. A further `us54`-specific rule bears on the economics: declares may be made at any time, including out of turn, so a teammate can beat you to a book you were saving [4].

3 The absorbing-containment theorem

Definition 1 *A book B is* contained *by team T in a state s if B is unresolved and every one of its six cards is held by a seat of T.*

Theorem 1 (Absorbing containment) *Let B be contained by team T. Then, under both `us54` and `pagat48`:*

(i) no seat of the opposing team has a legal ask into B;

(ii) any declare of B by a seat of the opposing team awards B to T; and

(iii) containment persists until T itself resolves B.

Proof. (i) Asking into B requires holding at least one card of B (row 6, identical in both rule sets); an opponent holds none.

(ii) A declare must assign all six cards to seats on the declarer's own team (row 12). Since every card of B sits with T, every assignment made by an opposing declarer is wrong. Under `us54`, row 14 awards a declare with any error to the opposing team — which is T. Under `pagat48` the rule that fires is likewise its row 14: at least one card of B is held by an opponent of the declarer (indeed all six are), so the opposing team — again T — scores the book. The `pagat48` void (its row 15) cannot apply: it requires the *declarer's own* team to hold all six, which contradicts containment.

(iii) The cards of B can move only through a hit on an ask into B or through resolution of B. By (i) opponents cannot ask into B at all. A holder's ask into B must name a card the holder lacks (row 7) at an opponent — who holds no card of B — so it can only miss, and a miss moves no cards (row 10). Hence every card of B stays with T until B is resolved, and by (ii) an opponent's attempt to resolve it awards it to T. The only transition out of the contained state that does not score for T is impossible; the state is absorbing. □

Corollary 1 (The turn-pass) *Let a seat of T hold k cards of a contained B, $1 \leq k \leq 5$. Then that seat may legally ask any chosen opponent for any card of B it does not hold (rows 5–7); the ask is a guaranteed miss, so the turn passes to the chosen opponent (row 10); the miss moves no cards, so the licence is never consumed and the move repeats indefinitely; and resolving B destroys the move.*

These are claims C1–C6 of the project's containment report [2], and their evidence tier there is *measured*, not *derived*: each is executed against the engine in `tests/engine/containment.test.ts` on every commit. In the pinned position (team A at seats $0, 2, 4$ containing LOW-C), the opponents have zero legal asks into the book (C1); an opponent's declare resolves with outcome `team0`, the book gifted back (C2); the holder's ask is legal, misses, and sends turn $0 \to 3$ as aimed (C3/C4); five consecutive uses leave the hand unchanged (C5); and after claiming, zero asks into the book remain (C6) [2]. The independent attestation is Develin's strategy text [7], which advises holding a fully controlled suit without declaring — the only way the opponents can win it is if you declare it wrong — and separately names asking into it as a way to pass information to teammates. The project's tier discipline explicitly refuses to count the recent cluster of AI-generated Literature repositories, including its own siblings, as corroboration [2].

3.1 The correction: `pagat48` is absorbing too, and a draft said otherwise

An earlier draft of the project's own documentation claimed that containment was *not* absorbing under `pagat48`, on the argument that its row 15 would void an opponent's declare of a contained book [3]. That is false, and it was caught by measurement rather than by re-reading: row 15's void requires the declarer's own team to hold all six cards, which cannot describe a seat whose opponents hold all six; the rule that actually fires is row 14, worded identically in both rule sets. Executed under `pagat48`, the opponent's declare of a contained book resolves with outcome `team0` and score $[1, 0]$ — gifted back, not voided — and the test suite now pins both C1 and C2 under the 48-card rules so the corrected fact cannot decay into folklore [3].[1]

The correction has a practical edge: the policy of Section 5.2 refuses to run under `pagat48`, and the refusal is now honestly labelled a *compatibility* gate — the project holds the shipped 48-card game byte-identical, verified by differential digest — rather than a rules gate. The mechanism is valid there and is declined anyway; no rules argument stands in the way of revisiting it if the 48-card roster is ever re-tuned [3].

4 The move and its information price

The incoming research handoff described the turn-pass as information-free. Measured, it is not — but the qualification is narrow and makes the move better, not worse [2]. Each ask publishes, via the public log (row 17), that the asker holds at least one card of the book and *lacks the named card*. The first time a given card is named, the asker's hand is narrowed by one card. Naming the *same* card again publishes nothing new, and the miss is equally guaranteed. The discipline that follows is the report's own: *reuse one card as your turn-pass*. The first use spends one card of hand information; every repeat is genuinely free; cycling through distinct cards leaks a bit each time for no gain. In the measured position the holder had four distinct nameable cards — four lifetimes of the move, if it were ever worth spending them [2].

Two framings of the same ask coexist and are deliberately kept apart [2, 3]. The rules-derivation frames it as *turn control*: the miss chooses who acts next. Develin frames it as *signalling*: the publication reaches the teammates, who — unlike the opponents, whom C1 and C2 lock out — can convert it, for instance by declaring the book, which C2 denies the other side. Both are modelled; only turn control decides (Section 5); signalling enters at exactly one term, the information price, which a `signalling` style books as delivered rather than spent.

Under `us54` the theorem's "no defensive value in claiming" meets one genuine countervailing cost that `pagat48` does not share: declares are legal at any time, so *a teammate may declare the contained book wrongly first*. That intra-team race — measured by the roster's `raceLosses` diagnostic — is the real price of waiting, and it binds the declare policy, not the ask policy [2].

5 The trigger, derived

Passing the turn is normally bad: you surrender tempo for nothing. The move is worth playing only when the ordinary alternative is poor *and* aiming the concession buys something. Both

[1] Prose decays faster than assertions. At the time of writing, one bot-level test in `tests/bots/contained.test.ts` still carries the superseded rationale in its name and comment ("row 14 gifts, row 15 voids") while asserting behaviour that remains correct for a different reason; the engine-level suite carries the correction. The episode is a working example of why this project records evidence tiers and executes its claims.

quantities are read off public data (row 17), in units of cards [3].

Write p^* for the hit probability of the ask the style would otherwise play, t_o for the opponent that ask concedes the turn to on a miss, and t_c for the opponent the style's aim (`missTarget`) would pick. Let $V_{\text{hit}} = 1$ card (the card taken; the asker also keeps the turn, row 9) and let $V_{\text{miss}}(t) = -\,\text{cost}(t)$, the cards a conceded turn is expected to hand seat t. Then

$$\begin{aligned} \text{value(ordinary)} &= p^*\,V_{\text{hit}} + (1-p^*)\,V_{\text{miss}}(t_o), \\ \text{value(pass)} &= V_{\text{miss}}(t_c) - \text{infoCost}, \end{aligned}$$

and the pass wins exactly when

$$p^* \;<\; \frac{a \cdot \text{gain} - \text{infoCost}}{\text{tempo}}, \tag{1}$$

with every term defined and sourced as follows [3]:

term	value	where it comes from
E	$\text{hits}/\max(1, \text{misses})$	over the public log. Row 9 keeps the turn on a hit and row 10 ends it on a miss, so a turn is a geometric run of hits with expected length $h/(1-h)$ — on counts, exactly hits over misses.
$\text{cost}(t)$	$E \cdot n_t/\bar{n}$	a conceded turn is worth more to a seat holding more ask licences (row 6); hand size n_t is the public part of that (row 17).
gain	$E \cdot (n_o - n_c)/\bar{n}$	what aiming the concession buys: $\text{cost}(t_o) - \text{cost}(t_c)$.
tempo	$1 + E \cdot n_o/\bar{n}$	the swing between hitting and missing: one card taken, plus the turn *not* conceded to t_o.
infoCost	0 on a reuse, else $1/U$	the Section 4 price, charged at the only channel C1 and C2 leave the opponents: count exhaustion.
a	the style's `containedPass`	the appetite (Section 5.1).

Here $\bar{n}$ is the mean hand size over seats still holding cards and U the number of live cards whose holder is not yet publicly certain. Every input is public and none is a tuned constant: an engine, a roster, or a rule set that hits more often prices a conceded turn higher, automatically. p^* is the seat's own *best* estimate — the constraint-refined probability where the skill model provides one, not the slot prior — so the ask being displaced is never undervalued. Two consequences fall out of the algebra, and they are why this is not "fire whenever legal" [3]:

- gain ≤ 0 whenever the ordinary ask already concedes where the style wants it, and the move is then refused however contained the book is. This switches the mechanism off wholesale for two styles with no bespoke code: Blitz aims at the *largest* hand (`missTarget 'most'`), so its ordinary concession is already its preferred one; a style with `missTarget 'random'` expresses no aim and buys nothing by aiming.
- A certain hit is never displaced: it is riskless material *and* keeps the turn (row 9). The threshold is clamped strictly below 1 and the check is explicit.

The implementation is deliberately small enough to quote. The valuation in `lib/engine/bots/contained.ts` is, in its entirety of consequence (lightly adapted: the file computes these same expressions and returns `threshold` as a field of its result object rather than assigning it):

```
const E = hits / Math.max(1, misses)
const meanHand = seats > 0 ? sum / seats : 0
const perCard = meanHand > 0 ? E / meanHand : 0
const gain  = perCard * (nOrd - nChosen)
const tempo = 1 + perCard * nOrd
const raw = (style.containedPass * gain - infoCost) / tempo
threshold = Math.max(0, Math.min(1 - 1e-9, raw))
```

One knob is deliberately *not* consulted: `minHitP`, the roster's long-shot filter. That knob refuses asks whose chance of taking a card does not justify the turn they risk; this ask has no chance of taking a card by construction and risks nothing, because the turn is the entire content of the move rather than its downside. Applying a hit-probability floor to it would refuse it always, at every style that carries one [9].

A worked instance, pinned by test [10]: seat 0 holds five of LOW-C with the sixth placed on a teammate by three public misses, opponents at 20/5/2 cards, and the log showing 3 hits to 3 misses, so $E = 1$ and $\bar{n} = 49/6$. Then gain $= 18/\bar{n} \approx 2.20$, tempo $= 1 + 20/\bar{n} \approx 3.45$, threshold ≈ 0.64: the pass fires. Add six filler misses so $E = 1/3$ and the same position prices at 0.404 for appetite 1 — below the ordinary ask's $p^* = 0.465$, so Balanced keeps its material ask — while the Hoarder's appetite of 1.33 lifts the bar to 0.538 and it spends the turn to aim the concession at the two-card seat. The same knob, the same arithmetic, two different styles.

5.1 The appetite: one number, eight of nine styles share it

Inequality (1) prices a *single* use, but the licence is renewable (C5): a style that will hold the book longer gets more uses out of the same first-use cost. The style's sole contribution is therefore `containedPass`, the *expected number of uses the licence gets before the book is banked*. A style that banks at its next opportunity takes 1; eight of the nine roster styles do. The Hoarder takes 1.33, read off the project's own measurement of how much longer it holds a set — `declareLatency` 31.02 window-cycles against Balanced's 23.39 — and it is the only appetite above break-even because it is the only delay that has actually been measured [3].

The uniformity is deliberate: which styles *reach* contained books, and whether aiming buys anything, are already governed by knobs the roster carried before (`hoardBooks`, `declareOnlyOwnHand`, `missTarget`), and nine per-style appetites would have hidden that behind nine tuned constants. An honest check on the number: measured uses per contained book run Turtle 3.40, Hoarder 2.64, Banker 2.56, Balanced 2.23, Punter 1.63 — the *ordering* the appetite argument predicts, but a Hoarder-to-Balanced ratio of 1.18, not the 1.33 the latency ratio implied. The number is right in sign and roughly right in size, and it has deliberately not been re-derived from 1.18, because fitting an appetite to the behaviour it produces is tuning against the outcome [3].

5.2 Implementation discipline

Three lines the implementation refuses to cross, each pinned by test rather than asserted [9, 10]:

Card reuse, stateless. The card chooser prefers a card of the book whose absence from this hand is already public; failing that it takes the canonical-first card the seat does not hold — which, because a contained book's cards can never move (Theorem 1(iii)), is fixed for the rest of the game, so every later call finds that same card already in the log. The reuse discipline holds with no state carried between decisions. Measured in play, 97.6% of the Turtle's uses and 94.8% of the Hoarder's cost no information at all [3].

No partner model. Develin records that prearranged conventions are forbidden in this tradition, and the containment report says not to blur the line [2, 7]. Signalling through a legal public ask is not a prearranged convention — but only as long as nothing decodes it. So nothing does: a teammate's knowledge state after a turn-pass is bit-for-bit the knowledge it builds after any other miss on the same card by the same seat — the row-17 public facts and nothing else — and the test suite asserts that equality directly, comparing the two knowledge objects. The inference layer does not know the ask was a turn-pass, and must not be taught to.

Gates before arithmetic. The plan is refused, cheapest check first, on: a zero appetite (every shipped difficulty tier has `containedPass` = 0, keeping the live product untouched); the rule set (the Section 3.1 compatibility gate); a skill that cannot read hand counts (a seat that cannot compare hand sizes cannot aim); a certain hit; then the recogniser; then Inequality (1). The recogniser's predicate is deliberately weaker than the report's measured position: it requires every card of the book to be *certainly on the seat's team*, not pinned to named seats — a pinned book is one the certain-claim path has already banked, so a pinning recogniser would fire only for the two styles that refuse certain declares. Under the weaker predicate the ask still cannot hit, and every style can reach the state [9].

6 Measurements

6.1 Instruments

Four instruments, in increasing order of scale, following the lab's duplicate-deal and fixed-precision protocol [3, 6]: (i) *decision mirrors*: 100 `us54` games per style in which two vectors identical but for `containedPass` are handed the same seat view and seed at every decision point and their chosen actions compared; (ii) *duplicate-deal pairs*: 1,200 pairs (2,400 games) per style, both orientations of every seed with a shared seat rotation, the pair as the unit of analysis; (iii) *matrix v2*: the full 36-cell round-robin at 4,300 pairs per cell (309,600 games, standard error ≤ 0.005 per cell) re-run on the same seed set as matrix v1, so the two runs differ by the policy change and nothing else — both artifacts committed (`src/lab/data/style-results.dominant.json`, records digest `087b626cedc5845e`, engine commit recorded `819eebb-dirty`; `style-results.v2.json`, digest `0c3cb524a3534d4a`, engine commit recorded `1667a1d-dirty` — the `-dirty` suffixes reflecting generation from a modified working tree at those commits); and (iv) *differential digests*: seeded games serialised action-by-action and hashed against the prior commit. The v2 health gate is clean: 0 illegal actions, 0 capped games, 0 invariant violations, 0 ties, 0 voids, 0 non-clinch terminations, 4,300 of 4,300 distinct seeds.

The digests bound the blast radius exactly [3]: under `pagat48`, both the three shipped tiers and all nine roster styles are bit-identical to the prior commit; under `us54` the shipped tiers are bit-identical (every preset carries `containedPass` = 0) and the roster changed for seven of nine styles — the two unchanged, Blitz and Scout, being exactly the two the model predicts (aim already at the largest hand; almost never holds a contained book). Even that count needed correcting by measurement: at 40 games the Archivist also digested as unchanged, because 40 games is below its firing rate; at 250 duplicate pairs (500 games) its digest differs, and 7-of-9 is the corrected figure.

6.2 Opportunity, fire, and refusal

Table 1 gives the decision-mirror rates. `opp` counts ask decisions at which a contained book was available at all; `fire` counts decisions at which the valuation returned a plan; in every style the number of divergent decisions equals `fire` exactly — the mechanism never once chose the move the ordinary policy was already going to play [3].

Table 1: Opportunity and fire rates, 100 mirror `us54` games per style [3].

style	appetite	aim	decisions	opp	opp %	fire	fire/opp	div %
Balanced	1	fewest	9,741	379	3.89%	29	7.7%	0.0414%
Blitz	1	most	9,345	472	5.05%	0	0.0%	0.0000%
Punter	1	fewest	9,507	347	3.65%	13	3.8%	0.0190%
Banker	1	fewest	9,818	495	5.04%	69	13.9%	0.0981%
Turtle	1	fewest	17,285	9,310	53.86%	679	7.3%	0.5575%
Hoarder	1.33	fewest	11,418	1,707	14.95%	211	12.4%	0.2580%
Scout	1	fewest	11,292	43	0.38%	0	0.0%	0.0000%
Ghost	1	fewest	9,996	131	1.31%	21	16.0%	0.0293%
Archivist	1	fewest	10,383	105	1.01%	5	4.8%	0.0067%

Three readings [3]. First, *the opportunity is a property of the declare policy, not the ask policy*: a contained book exists at 53.86% of the Turtle's ask decisions (it never banks a set split across teammates, so it accumulates them) and 0.38% of the Scout's (whose information weighting pins holders early, and a pinned contained set is banked immediately). The Hoarder sits second at 14.95% — nearly four times the control's 3.89% — which is precisely the benefit the project once said it was paying for and not collecting. Second, *it is mostly a refusal*: even where the licence exists it is spent at 3.8–16.0%, and every refusal is the gain ≤ 0 arm — the ordinary ask was already conceding where the style wanted. Third, *divergence is small*: the largest mover is the Turtle at 0.56% of all decisions; seven of nine move less than 0.1%; exactly two — Blitz and Scout — do not move at all (the Archivist fires five times, 0.0067%). The information price, for completeness: charged only on first uses (2.4% of the Turtle's, 5.2% of the Hoarder's), bounded by $1/U < 0.05$ cards in these positions against a tempo above 1, and it never flipped a decision.

6.3 Does it win? No.

Table 2 is the head-to-head: each style *with* the mechanism against the identical style with it forced off, 1,200 duplicate-deal pairs each; 0.5 is no effect; `changed` counts games whose result or length differed from the same deal played by two copies of the unequipped vector — games the mechanism actually touched [3].

No style is two standard errors from 0.5. The largest deviation in either direction is 1.6 SE (Ghost, 0.4958 ± 0.0027); the two heaviest users land on and slightly below the control (Turtle 0.5008 ± 0.0065, Hoarder 0.4938 ± 0.0054); across the seven styles that fire at all, point estimates straddle 0.5 (four below, three above, mean 0.4991). Blitz and Scout return exactly 0.5000 ± 0.0000 over 2,400 games each, which is the zero-changed-games column restated: for those two, the vectors are the same player [3].

Table 2: Per-style null result: WITH vs. WITHOUT `containedPass`, 1,200 duplicate-deal pairs (2,400 games) per style [3].

style	appetite	pairs (games)	WITH score	SE	changed
Balanced	1	1,200 (2,400)	0.5046	0.0034	239
Blitz	1	1,200 (2,400)	0.5000	0.0000	0
Punter	1	1,200 (2,400)	0.4971	0.0029	142
Banker	1	1,200 (2,400)	0.5021	0.0035	281
Turtle	1	1,200 (2,400)	0.5008	0.0065	1,268
Hoarder	1.33	1,200 (2,400)	0.4938	0.0054	915
Scout	1	1,200 (2,400)	0.5000	0.0000	0
Ghost	1	1,200 (2,400)	0.4958	0.0027	156
Archivist	1	1,200 (2,400)	0.4996	0.0004	11

6.4 The full matrix confirms it

The mirror isolates the mechanism against a twin; matrix v2 exposes it to the whole roster. Nothing that matters moved [3]: the verdict is unchanged (dominant style: Punter, all four criteria); the ranking is identical position for position; Punter's mean score is 0.5829 in both runs and its maximin 0.5193 $\to$ 0.5190 (its worst matchup relabelled between two cells that were already tied); cyclic energy rose 0.0097 $\to$ 0.0112 against a threshold of 0.15, with the same single directed 3-cycle (Hoarder > Archivist > Ghost) still failing Benjamini–Hochberg (q 0.31 $\to$ 0.27); significant cells went 34 $\to$ 33 of 36, entirely because one marginal Hoarder cell fell out of significance; Nash and α-Rank both still put all mass on Punter. No cell moved significantly: the largest paired per-deal difference is `blitz-vs-ghost` at $+0.0030 \pm 0.0015$ ($t = 2.07$), which does not survive Benjamini–Hochberg over 36 comparisons — one $|t|$ above 2 in 36 is what chance produces. Per style, paired over the same 4,300 deals, every mean score is within 1.5 SE of zero change (Table 3); the headline row is the Hoarder — the style built to exploit the mechanism — at $\Delta = -0.0018 \pm 0.0014$ ($t = -1.35$), sixth of nine in both runs.

Table 3: Matrix v2 vs. v1: per-style mean score, paired over the same 4,300 deals per cell [3, 11].

style	v1	v2	Δ	SE(Δ)	t	rank
Punter	0.5829	0.5829	−0.0001	0.0009	−0.07	1 → 1
Blitz	0.5594	0.5602	+0.0008	0.0008	+0.94	2 → 2
Balanced	0.5584	0.5581	−0.0003	0.0009	−0.34	3 → 3
Banker	0.5274	0.5285	+0.0011	0.0011	+1.00	4 → 4
Ghost	0.4928	0.4928	+0.0000	0.0010	+0.03	5 → 5
Hoarder	0.4904	0.4885	−0.0018	0.0014	−1.35	6 → 6
Archivist	0.4791	0.4802	+0.0011	0.0008	+1.47	7 → 7
Scout	0.4064	0.4063	−0.0001	0.0007	−0.12	8 → 8
Turtle	0.4032	0.4025	−0.0007	0.0019	−0.38	9 → 9

Two rows of Table 3 are not evidence of anything and should not be read as such: Blitz's and Scout's deltas are nonzero even though their own policies are byte-identical between runs, because their *opponents* changed [3].

6.5 Exploitability, and the axis the search cannot turn

The single-knob exploitability search was re-run from scratch for all nine styles (its cache keys on a hash of the engine tree, so the new file invalidated every entry). The deltas are all inside the search's own noise — its final measurements carry standard errors of 0.008–0.017 and it only accepts moves worth at least ≈ 0.03 — and the verdict criterion passes in both runs, with the dominant style's exploitability moving $0.0000 \to 0.0025$ against a rivals' median of 0.0444 [3]. The one qualitatively interesting row is the Hoarder's: the best single-knob counter to the Hoarder-with-the-benefit is no longer a scoring weight but `hoardBooks=0 minHandSize=0` — *stop hoarding* — worth 0.5563 ± 0.0124. The counter to the style is still to not be it [3, 11].

One gap remained: the search's knob ladder predates `containedPass`, so neither run could switch the mechanism itself on or off. That axis was probed directly with the same sequential protocol used for tuning (SPRT, $d_0 = 0$, $d_1 = 0.03$, $\alpha = \beta = 0.05$, 24–400 pairs, then a fixed-N 400-pair evaluation on unseen seeds), against every style, at appetites $\{0, 2, 4\}$. **No candidate was accepted, for any style, at any appetite.** The two rows worth naming: turning the mechanism *off* against the Turtle ran to the pair cap unresolved ($+0.0125 \pm 0.0120$ in search; 0.5225 ± 0.0121 in eval) — the one candidate the SPRT could not decide; and the same candidate against the Hoarder produced a sign disagreement between search and eval (-0.0200 ± 0.0176 rejected, versus 0.5275 ± 0.0098), recorded precisely because it is the one number in the exercise that would have looked like a finding if only its second half had been reported — the honest summary of two contradictory 0.01–0.02-sized effects is that neither is real. What *does* resolve is over-use: at appetite 4 the Turtle drops to 0.4600 ± 0.0148 and the Banker to 0.4850 ± 0.0083. The move is worth nothing, and spending turns on it anyway is worth less than nothing [3].

7 Why the effect is small

The null is not mysterious; the trigger's own terms bound it. A conceded turn is worth $E \cdot n_t/\bar{n}$ cards, and the measured E — hits over misses on the public log — is around 1 at this roster's level of play. So the *entire* spread between the best and the worst opponent to concede to is on the order of one card, and the aimable fraction of that spread is all the move can ever buy. Against it stands everything the ordinary ask gives up: p^* of a card of material, plus the turn a hit would have retained (row 9). The move is real, and it is small [3]. The firing profile of Table 1 says the same thing from the other side: most of the time the concession is already aimed correctly, so there is nothing to buy, and the valuation — honestly computed — mostly refuses.

Under `us54` there is also an active cost to the holding pattern the move requires: the intra-team race. A contained book left unclaimed is a book a teammate can declare wrongly at any time, gifting it away (rows 11 and 14 of [4]); the Hoarder's `raceLosses` doubles ($0.046 \to 0.091$ per game) when its hoarding is actually expressed [3]. The absorbing state is safe from the opponents, not from one's own side.

8 What is not settled

The declare-policy mechanism. Everything measured here changes the *ask* policy: when a style already holds a contained book, it may spend a turn aiming a concession. The containment report's second recommendation — hold contained books by default, trading tempo and race risk against the retained licence, instead of treating banking as automatically correct — is a change to the *declare* policy and is untouched [2, 3]. Table 1's first reading is the evidence that this is where the leverage would be: the opportunity rate is set almost entirely by how long a style leaves books unclaimed (53.86% for the Turtle against 0.38% for the Scout), i.e. by the declare policy. The report's expected-value analysis is on file for whoever attempts it: with $c = P(\text{contained})$ and $a = P(\text{assignment correct} \mid \text{contained})$, declare iff $c > 1/(1+a)$ under `pagat48`'s void rule and $c \cdot a > 1/2$ under `us54`'s gift rule — and do not collapse c and a into one number, as the current claim planner does, since "is this book on my team" and "do I know which teammate holds each card" have different distributions and different remedies, and conflating them biases the bot conservative [2].

An untuned roster. The bar Inequality (1) must clear is *the best ordinary ask the style has*, and the project's tuning protocol has never been run over this roster; a roster with a weaker ask policy would leave more room for a turn-pass [3]. Relatedly, the deliberate known-miss ask has been independently flagged as a gap: the FishBot v0.3 paper's engine also permits known-miss asks and its policy "almost never" chooses them [2]. Two independent sources plus a measured mechanism made the gap real; this paper measures what closing it is worth at one operating point, not at all of them.

9 Threats to validity

The claims are about this rule set as implemented. C1–C6 are measured against this engine; that is the strongest tier available here, and also its scope. The independent attestation (Develin, read directly) covers the absorbing property and both framings of the ask, not the quantitative results [2].

Self-play, one roster, one engine. The null is measured between policy vectors sharing one inference engine, on a nine-style roster at one level of play. A stronger or weaker population could price the move differently; Section 8 names the direction.

The appetite is a proxy. `containedPass` $= 1.33$ is read off a latency ratio, and the realised use ratio is 1.18. The discrepancy is reported and deliberately left unfitted [3].

Search noise. The exploitability deltas of Section 6.5 sit inside the search's own error bars, and the one sign-contradictory result is reported as noise, not evidence. The SPRT probe of the `containedPass` axis closes the ladder gap but inherits the same detection floor (≈ 0.03); effects smaller than that would not have been accepted — though the paired mirror of Table 2, with standard errors of 0.003–0.007, would have seen them.

Small-sample traps, twice. Forty games misread the Archivist's digest as unchanged; a 200-pair pilot misplaced the Hoarder's rank (7th, against 6th in both full-precision runs). Both errors were caught by re-measuring at scale and both corrections are recorded in the source documents [3]. They are kept in this paper as calibration for how much to trust any single small run — including the ones reported here.

10 Conclusion

A theorem can be true, its mechanism implementable, its implementation faithful, and the whole thing worth nothing. The contained book is absorbing under both rule sets this project supports — a fact strong enough to survive the project's own attempt to disprove half of it — and the turn-pass it licenses is renewable, aimable, and after its first use free. Implemented with an honest price and handed to the style built to exploit it, it changed 915 of that style's 2,400 mirror games and moved its score by -0.0062 ± 0.0054 against its twin and -0.0018 ± 0.0014 against the field. The measured value of the move at this roster's level of play is zero, and the arithmetic that prices it says why: a conceded turn is worth about one card here, and one card, aimed or not, does not win races to five.

The negative result is the deliverable. It converts "the Hoarder underperforms because its benefit was never implemented" — a live objection, on the record — into "the benefit is implemented, reached nearly four times as often as baseline, priced above break-even, and worth nothing," and it does so with the objection's own instruments. What remains open is exactly one mechanism (the declare-side holding policy), and the measurements here say where its leverage would have to come from. Measuring rather than asserting cost two corrections on the way to a zero; both corrections, and the zero, are the point.

References

[1] A. Hsieh, *FishAI v0.5: measuring play style without skill in a 54-card Literature variant*, companion paper, published alongside this one, 2026. The engine, the nine-style roster, the tournament design, and the full-matrix results this paper's measurements plug into.

[2] A. Hsieh, *CONTAINMENT.md — the contained-book result*, FishAI project internal report, August 2026. Claims C1–C6 with evidence tiers, the information-cost correction (§1.2), the Develin attestation (§2), and the claim-EV thresholds (§3.3).

[3] A. Hsieh, *STYLES.md — the play-style roster under* `us54`, FishAI project internal report, August 2026. §6.2 (the Hoarder prediction and its refutation), §6.3 (the turn-pass implemented and measured; §6.3.7 the `pagat48` correction), §6.4 (matrix v2).

[4] A. Hsieh, *RULES_US54.md — the "US student" 54-card variant*, FishAI project internal report, August 2026. Decision-table rows cited by number throughout.

[5] FishAI project, *RULES.md — the 48-card pagat baseline*, internal report, 2026. Rows 14–15 (the split declare-error outcomes, including the void).

[6] FishAI project, *BOT_LAB.md — simulation-lab protocol*, internal report, 2026. Duplicate-deal pairing (§5.1–5.2), fixed-N matrix precision (§5.4), SPRT tuning (§5.5), and the exploitability ladder (§5.7).

[7] M. Develin, *The Ten Best Card Games You've Never Heard Of*, ch. 9 (Canadian Fish), `http://www.bantha.org/~develin/cardgames.html`. Independent attestation of the absorbing property and of the signalling framing; records the tradition's ban on prearranged conventions.

[8] J. McLeod (ed.), *Literature (Canadian Fish)*, Pagat card-game rules archive, `https://www.pagat.com/quartet/literature.html`. Folklore source for both rule sets' common core.

[9] FishAI project, `lib/engine/bots/contained.ts`, the turn-pass policy: recogniser, card reuse, valuation, and gates. The mechanism claims are pinned by `tests/engine/containment.test.ts` (C1–C6 under `us54`; C1–C2 under `pagat48`; an end-to-end policy-through-reducer step).

[10] FishAI project, `tests/bots/contained.test.ts`, the decision-level pins: the refusal on a level board, the measured-E worked positions, the certain-hit clamp, the appetite's scaling, the no-partner-model equality, and the shipped tiers' zero appetite.

[11] FishAI project, committed measurement artifacts `src/lab/data/style-results.dominant.json` (matrix v1, engine commit recorded `819eebb-dirty`, records digest `087b626cedc5845e`) and `src/lab/data/style-results.v2.json` (matrix v2, engine commit recorded `1667a1d-dirty`, records digest `0c3cb524a3534d4a`); 309,600 games each, health gates clean.